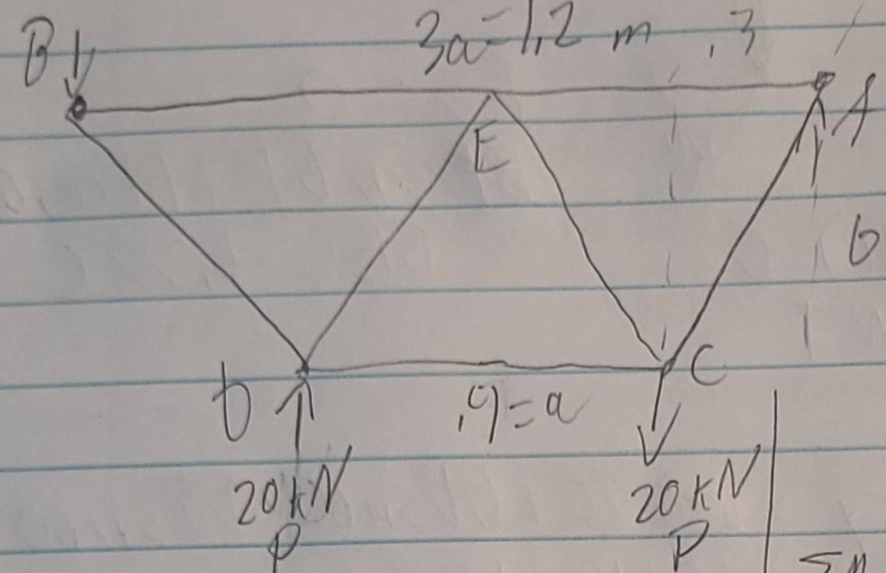
$1.2/2 = .6/2 = .3$




Outside forces:

$\Sigma F_x = A_x = 0$

$\Sigma F_y = A_y + B_y = 0$

~~$\Sigma M_A = +20 kN(\ldots) + 20 kN(\ldots)$~~

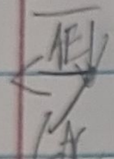
$\Sigma M_A = (-20kN(.3m)) + 20kN(.9m) - B_y(1.2m) = 0$

$B_y = 10kN$

$\Sigma F_y = A_y + 10kN = 0$

$A_y = -10kN$

A: $\Sigma F_y = A_y + AC\sin 60 = 0$

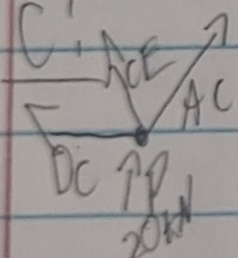



$AC = +11.547\ C$

$\Sigma F_x = +11.547(\cos 60) + AE = 0$

$AE = 5.77$

C:




~~$\Sigma F_y = AC\sin 60 + EC\sin 60 + 20kN = 0$~~

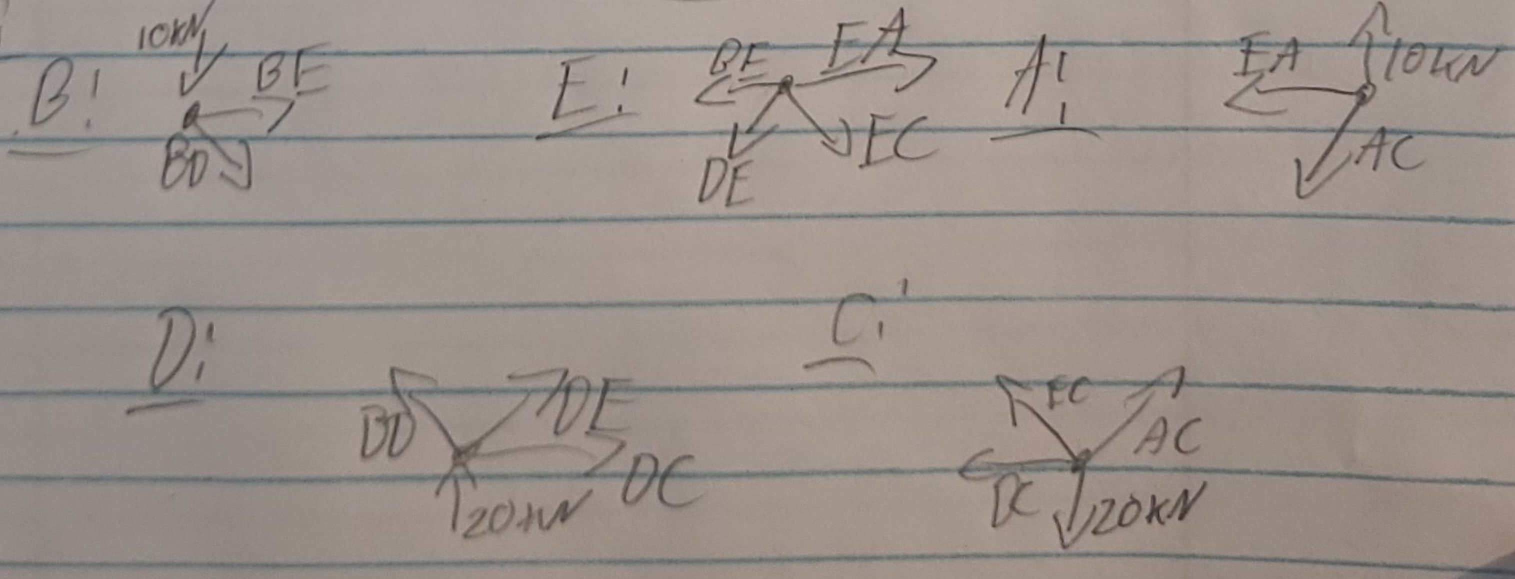
~~$\frac{-20}{\sin 60}$~~

$F_y = 20kN + AC\sin 60 + EC\sin 60 = 0$

$20 + 10 + EC(.866) = 0$

$EC = 34.64\ kN$ compression

FBD:



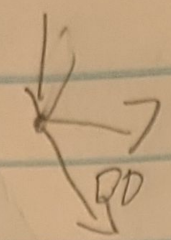
B

$F_y = 0$

$B_y - F_{BD}\sin 60 = 0$

$10\,kN - BD(.866) = 0$

$BD = 11.55\,kN$ T

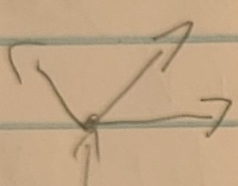



$F_x = 0$

$BE + BD\cos 60 = 0$

$BE = -5.77$ C

D



$F_y = 20 + BD\sin 60 + DE\sin 60 = 0$

$20 + 10 + DE(.866) = 0$

$DE = 34.64$ C

$F_x = -BD\cos 60 + DE\cos 60 + DC = 0$

$-5.77 + 5.77 + DC = 0$ $DC = 0$